The Discrete Perception of Continuous Speech

Spencer Caplan (CUNY Graduate Center) & Karthik Durvasula (Michigan State University)
scaplan@gc.cuny.edu

The fact that speech perception is well modeled by a sigmoid curve is perhaps the most famous fact in the history of the psycholinguistics of speech. Yet, incrementally varied stimuli often elicit graded behavioral responses, a pattern commonly taken as evidence that perceptual representations themselves are continuous [1-2]. This inference from graded behavior, however, conflates the behavior itself with the nature of the underlying mental representations and processes which generate such behavior. Smooth-clined response curves—particularly when averaged across trials or participants—do not uniquely diagnose the nature of perceptual or linguistic encoding [3]. Here we challenge the assumption of gradient representations and synthesize the surrounding debate by introducing the **Discrete Sampling (DS)** framework, a novel theory—alongside a computational model instantiating it— under which listeners map continuous phonetic input into discrete, category-based internal states via a biologically plausible sampling procedure.

Under DS, perception proceeds by repeatedly sampling discrete "labels" from the input relative to a noisy category boundary as speech unfolds (in line with e.g., [4]). Each simulated trial yields a perceptual encoding composed of a small number of such labels (e.g., /d/ or /t/), reflecting moment-to-moment variability in the category threshold applied (Fig 1a). Uncertainty is thus represented not as a continuous probability distribution but as a simple multiset of finite counts. These discrete representations can, in turn, be used to generate task-relevant linguistic inferences (i.e., lexical activation) and behavioral responses via a "resampling procedure" that derives both binary identification/discrimination judgements (Fig 1b) and scalar category-likeness ratings (Fig 1c) from the same perceptual encoding, while simultaneously yielding a natural measure of processing time.

We evaluate these a priori predictions to compare DS via computational simulation against a broad class of continuous alternatives using trial-level empirical behavioral data from 575 participants across three speech perception paradigms [4-6]. DS derives the classic sigmoidal curves. Furthermore, it derives the characteristic "upside-down U shaped" reaction-time profile (dating back to [8]) with slowest processing specifically at the category boundary. Critically, when listeners provide responses on a continuous scale (as in a VAS task) DS uniquely predicts the perseverance of this concave-down RT trend together with the in-tandem modulation of categorization curve slope and relative magnitude of the RT peak (Figs 2-3). Across paradigms, DS provides a unified account of categorization slopes, reaction-time dynamics, and variability patterns. In addition, DS captures systematic population-level structure in response behavior and inter-subject variability, which we omit here for space. Continuous models can capture graded choice behavior but systematically fail to account for the observed relationship between uncertainty, reaction-time profile, and task demands. In contrast, DS accurately predicts the observed relationships using a single representational framework.

These findings challenge the standard inference from graded behavior to continuous perceptual representations. DS offers a natural explanation of these results without encoding continuous acoustic values, storing and updating probability distributions, or fitting task-specific parameters. Rather, this suggests speech perception (and perhaps linguistic encoding in general) operates over a limited set of discrete internal states, with apparent gradience emerging from an online sampling procedure and decision dynamics rather than representational gradience. More broadly, DS offers a general framework for understanding how perceptual systems can encode uncertainty and support flexible behavior using limited computational resources, with implications for theories of linguistic knowledge/competence and processing more generally.

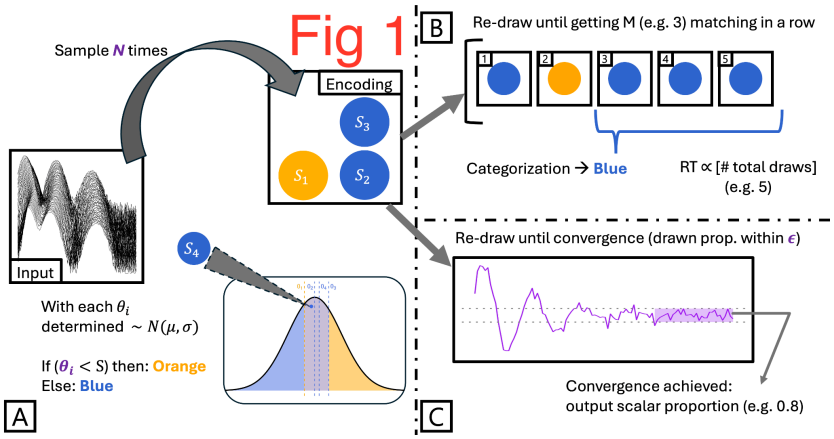


Fig 1: Schematization of Discrete Sampling (DS) architecture. (A) Perceptual encodings are constructed by a limited number of discrete category labels relative to a noisy boundary threshold. (B) Categorical responses are derived via resampling from the encoding until convergence; yielding both a response and a natural measure of RT (C) The same resampling dynamics support graded scalar responses.

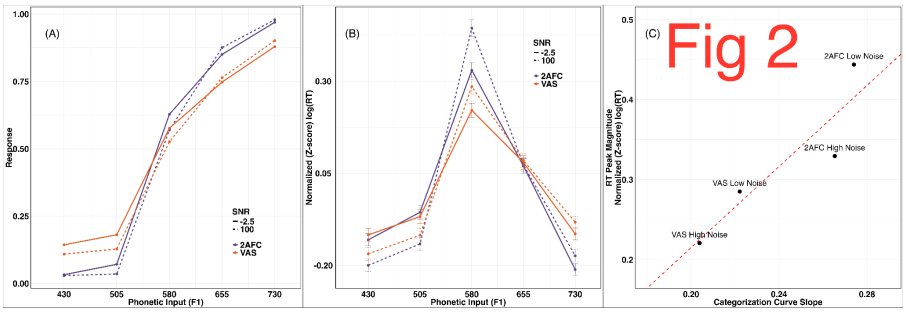
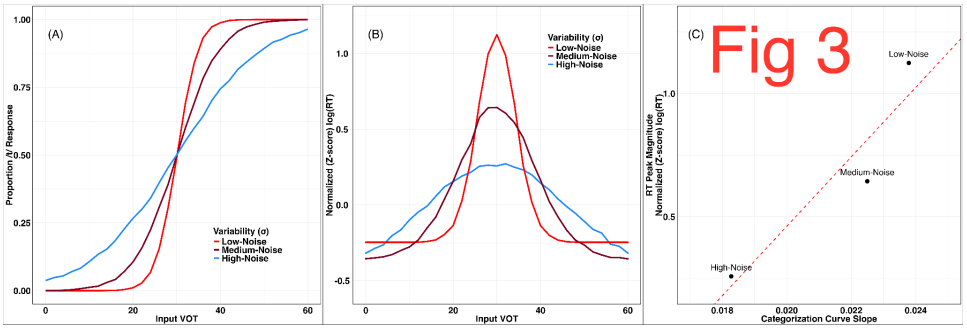


Fig 2: Empirical data from [7]. Categorization (left) and RT (middle) curves as a function of condition (which varied response type and signal-to-noise ratio). The right panel shows the systematic relationship between categorization slope and relative RT peak magnitude as a function of condition.

Fig 3: shows the same configuration as Fig 2 but derived via simulation under DS rather than human empirical trials



References: [1] Massaro & Cohen, (1983) *Categorical or continuous speech perception: A new test*, [2] McMurray (2022) *The myth of categorical perception*, [3] Estes (1956) *The problem of inference from curves based on group data* [4] Oderbolz et al. (2025) *Asymmetric sampling in time: Evidence and perspectives* [5] Caplan et al. (2021) *Now you hear me, later you don't: The immediacy of linguistic computation and the representation of speech* [6] Kleinschmidt et al. (2015) *Supervised and unsupervised learning in phonetic adaptation* [7] Rizzi & Bidelman (2024) *Functional benefits of continuous vs. categorical listening strategies on the neural encoding and perception of noise-degraded speech* [8] Pisoni & Tash (1974) *Reaction times to comparisons within and across phonetic categories*